Pranil Sakpal

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Education

Pimpri Chinchwad College of Engineering (PCCOE) **Pune, India**

Bachelor of Technology in Computer Engineering *Aug. 2023 – June 2027*

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, DBMS, Operating Systems, Object Oriented Programming

Technical Skills

Languages: C, C++, Python, R, SQL, HTML, CSS, Javascript, React.js, Node.js
Databases & Backend: Schema Design, MongoDB, MySQL
AI/ML: TensorFlow, Keras, scikit-learn, Numpy, Pandas, Computer Vision, Neural Networks, NLP, LLMs
Tools: Git, GitHub, Linux (Ubuntu), AWS, VS Code, Figma, Google Colab, PowerBI

Projects

Crop Recommendation System | *Python, Flask, Decision Tree, KNN, K-Means, ML* **Source Code**

- Built a multi-model ML pipeline using **Decision Tree, KNN, and K-Means** trained on **2,200+ samples** across **22 crop categories** to recommend optimal crops based on soil and climate features.
- Achieved **97%+ classification accuracy** with Decision Tree and KNN models, validated using train-test split and label encoding with StandardScaler normalization.

Eye Strain Detection System | *Python, Deep Learning, OpenCV, TensorFlow/Keras, Computer Vision* **Source Code**

- Implemented **CNN-based eye state classification** (open/closed) combined with **Eye Aspect Ratio (EAR)** for robust blink detection and fatigue analysis.
- Achieved **90% classification accuracy** on eye state detection and computed real-time blink rate metrics for fatigue assessment.
- Designed an automated **alert system** for low blink rate and prolonged eye closure, reducing potential eye strain risk by **30–40%**.

Research Experience

Undergraduate Research – Medical Imaging (ROP Detection) **Pune, India**

- Conducted research on automated detection of Retinopathy of Prematurity (ROP) using retinal fundus image datasets.
- Applied **CNN-based models** with advanced preprocessing techniques (CLAHE, normalization), achieving strong diagnostic performance in medical imaging tasks.

Experience

UI/UX Intern – PISTU | *Figma, Prototyping, Wireframing, Design* **May 2025 – Jul 2025**

- Designed **high-fidelity wireframes** and interactive **10+** prototypes in Figma, improving user flow efficiency and visual consistency.
- Collaborated with cross-functional teams in a **hybrid** work environment to deliver production-ready UI components within deadlines.

Executive Officer – International Relations Cell, PCCOE | *Management, Communication* **Aug 2025 – Present**

- Led and managed a **core team** for cross-cultural events including an international hackathon, Japan, and Germany programs.
- Enhancing PCCOE's global engagement by managing communication with **10+** international partner institutions and organizing **5+** international events.

Achievements

- **OpenCV Course:** Completed OpenCV computer vision course covering **15+** image processing techniques including object detection, contour analysis, and real-time video processing using Python.
- **IIT Bombay – Generative AI Workshop:** Completed intensive **2-day** Generative AI workshop at IIT Bombay Techfest covering hands-on model training and large language model fine-tuning methodologies.
- **Project Copyright:** Co-built **Government of India copyrighted** AI mental health chatbot leveraging **LLaMA 3 (70B)**, LangChain, React, and MongoDB Atlas to deliver judgment-free student support through real-time chat via Socket.io.
- Won **1st place** among **100+** participants in the MasterChef UI/UX design competition at Anantya 2026 (GDGC PCCOE), designing an innovative, user-centered interface in Figma.